Numerical Experiments for the Joint Score of Diffusion Models on Correlated Sequences

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Abstract

This document is a self-contained catalogue of every numerical experiment performed in the project on score-based diffusion of correlated sequences. For each experiment we record (i) the model and parameters, (ii) the quantity computed, (iii) the script and figure that produced it, and (iv) its current verification status. The experiments fall into five layers, listed here in order of foundational dependence rather than chronological order: a numerical-identity audit suite, the Gaussian AR(1) benchmark figure pack, the Laplace AR(1) K=1 suite, the Laplace AR(1) K=2 suite (with a proxy caveat), and the Conjecture 15 Hessian-rank sweep (with a regime-dependent caveat). Cross-references to the synthesis manuscript `merged_core_notes.tex` are given throughout.

Codebase root: `Score_Diffusion/`

Status (compile date): all experiments runnable from the scripts listed; paper-grade Conjecture 15 sweep pending HPC.

Contents

1 Setup, notation, and what we mean by "the score"

We consider a clean Markov chain $\boldsymbol{a} = (a_0, \dots, a_K)$ drawn from a prior $P_0(\boldsymbol{a}) = p_0(a_0) \prod_{u=0}^{K-1} M(a_{u+1} \mid a_u)$, each frame independently noised by an OU channel:

$$x_u = e^{-t}\, a_u + \sqrt{\Delta_t}\, \xi_u, \qquad \xi_u \sim \mathcal{N}(0,1) \ \text{ i.i.d.}, \ \Delta_t = 1 - e^{-2t}.$$

The *joint* score and Hessian of the noised marginal $P_t(\boldsymbol{x}) = \int P_0(\boldsymbol{a}) \prod_u \mathcal{N}(x_u, e^{-t}a_u, \Delta_t)\, d\boldsymbol{a}$ are

$$\mathcal{S}(\boldsymbol{x}, t) \;=\; \nabla_{\boldsymbol{x}} \log P_t(\boldsymbol{x}), \qquad \mathcal{H}(\boldsymbol{x}, t) \;=\; -\nabla^2_{\boldsymbol{x}} \log P_t(\boldsymbol{x}).$$

The experiments below probe the structure of $\mathcal{S}$ and $\mathcal{H}$ as t varies, never the per-frame marginal score $\nabla_{x_u} \log p_t(x_u)$. The distinction is the conceptual correction of the Mézard meeting (12 March 2026); it is documented in `CLAUDE.md` and in `merged_core_notes.tex` §2.3, and is the reason the older toy-model HTML demos (`Toy-models/Toy-model_1…6`) are not reused below.

Project naming convention. In `Experiments/scores_exact.py` the class names refer to the number of *innovations*, not frames: `LaplaceAR1_K1` is the 2-frame chain (a_0, a_1) with a single Laplace innovation, and `LaplaceAR1_K2` is the 3-frame chain (a_0, a_1, a_2) with two innovations. We follow this convention throughout.

2 Numerical-identity audit (foundational)

The audit suite verifies, to floating-point accuracy, every closed-form identity that the rest of the project relies on. It must be run first because the figure scripts and the `LaplaceAR1_K2` Fourier inversion both reuse the same building blocks.

Source.

- Driver: `Research/laplace_ar1_audit/code/run_checks.py`
- Library: `Research/laplace_ar1_audit/code/ar1_diffusion_utils.py`
- Output: `Research/laplace_ar1_audit/code/check_report.txt`

Method. Each check fixes a small set of parameters, evaluates the closed-form expression and an independent reference (matrix inverse, Neumann series, `scipy.integrate.quad`, or central finite differences), and reports the maximum absolute error.

Table 1: Audit suite errors, generated by `run_checks.py`.

Identity	Reference used	Max abs. error
Gaussian AR(1) chain inverse, $\Sigma_0\Sigma_0^{-1} = I$, n=9	matrix product	1.33×10^{-16}
Small-time Neumann expansion of Σ_t^{-1}, t=0.01, 20 terms	exact Σ_t^{-1}	8.88×10^{-16}
Deterministic-control inverse, $\Sigma_t\, \Sigma_t^{-1} = I$	matrix product	4.78×10^{-16}
Normal–Laplace convolution closed form, 9 test points	`scipy.quad`	1.11×10^{-16}
Laplace K=1 density (1D reduction) vs. direct 2D quadrature	`scipy.dblquad`	2.78×10^{-17}
Laplace K=1 score vs. central finite diff. (h=2×10^{-4})	FD	1.27×10^{-9}
Laplace K=1 Hessian vs. central finite diff. (h=2×10^{-4})	FD	1.91×10^{-8}

Remark. The first five rows are at machine precision; the last two are limited by the order of the finite-difference truncation error ($O(h^2)$ on $\log p$). Both are well below the resolution of any downstream figure.

3 Gaussian AR(1) benchmark

Source.

- Driver: `Research/research_continuation_v2/code/generate_all_figures.py`
- Library: `Research/research_continuation_v2/code/laplace_ar1_utils.py`
- Figures (PDF): `Research/research_continuation_v2/figures/fig01_*…fig11_*.pdf`
- Run summary: `Research/research_continuation_v2/code/figure_summary.txt`

Model. Stationary scalar AR(1): $a_{u+1} = \alpha a_u + \eta_u,\ \eta_u \sim \mathcal{N}(0, \sigma_\eta^2)$, of length $N = 60$, with

$$\alpha = 0.90, \quad \sigma_0^2 = 1, \quad \sigma_\eta^2 = 1 - \alpha^2 = 0.19.$$

At $t = 0$ the precision is exactly tridiagonal, $J_0 = \Sigma_0^{-1}$ with diagonal $(1+\alpha^2)/\sigma_\eta^2$ (interior) and off-diagonal $-\alpha/\sigma_\eta^2$ (modulo boundary corrections).

Quantities recorded across t**.** $\Sigma_t = e^{-2t}\Sigma_0 + \Delta_t I$ and $J_t = \Sigma_t^{-1}$; spectrum of Σ_t; tridiagonal mass fraction $\rho_3(t) = \sum_{|i-j|\le 1} |J_t|_{ij} / \sum |J_t|_{ij}$; effective bandwidth $b_{0.95}(t)$; mean precision-interaction range $\bar{r}(t)$; neighbour score correlation $\mathrm{corr}(\mathcal{S}_k, \mathcal{S}_{k+1})$; posterior-gain row $(e^{-t}\Sigma_0 J_t)_{m,\cdot}$, $m = N/2$; deterministic-control counterpart of J_t.

Figure pack. The eleven panels are:

Two-line takeaway. For Gaussian AR(1) the joint score is *exactly* linear, $\mathcal{S}(\boldsymbol{x}, t) = -J_t(\boldsymbol{x} - e^{-t}\boldsymbol{\mu}_a)$; locality of the score is therefore equivalent to sparsity of J_t, and Table 2 shows that this sparsity is broken for intermediate t and gradually restored only as $t \to 0$ or $t \to \infty$. A duplicate older figure set lives in `Research/anatomy_sigma_t/figures/` and `Research/sigma_t_analysis/figures/`; the v2 set above is canonical.

4 Laplace AR(1) at K=1 (warm-up)

Goal. Replace the Gaussian innovation $\eta_u \sim \mathcal{N}(0, \sigma_\eta^2)$ by a heavier-tailed Laplace innovation $\eta_u \sim \mathrm{Laplace}(0, b)$ of equal variance ($2b^2 = \sigma_\eta^2$), and compare the score and Hessian of the noised joint distribution to the Gaussian benchmark of Section 3. With $\alpha = 0.9$, variance matching gives $b = \sqrt{(1-\alpha^2)/2} = 0.3082$.

Source A — main figure pack.

- Driver: `Research/research_continuation_v2/code/generate_all_figures.py` (functions for fig12–fig14)
- Figures: `Research/research_continuation_v2/figures/fig12_*…fig14_*.pdf`

Source B — audit illustrations.

- Driver: `Research/laplace_ar1_audit/code/make_figures.py`
- Library: `Research/laplace_ar1_audit/code/ar1_diffusion_utils.py`
- Figures: `Research/laplace_ar1_audit/figures/fig01_*…fig04_*.pdf`

Figures.

File	What it shows
fig01_sigma0_vs_precision0.pdf	Side-by-side heatmaps of Σ_0 (dense, geometric decay) and J_0 (tridiagonal). Visual proof that the Markov property survives in J_0 at t=0.
fig02_sigma_t_panel.pdf	Five-column time series of Σ_t vs J_t for $t \in \{0.01, 0.15, 0.4, 1.0, 2.5\}$. Off-band fill-in of J_t is visible already at $t = 0.15$.
fig03_eigenvalue_trajectories.pdf	All 60 eigenvalues of Σ_t and of J_t on $t \in [0, 4]$; every eigenvalue tends to 1 (resp. 1) as $t \to \infty$.
fig04_tridiag_fraction_condition.pdf	$\rho_3(t)$ and the off-band complement on $t \in (0, 4]$, plus $\kappa(\Sigma_t)$ on a log axis. ρ_3 reaches its minimum ≈ 0.76 and the condition number explodes near $t \to 0^+$.
fig05_precision_row_profile.pdf	Row $m = 30$ of J_t vs frame index for six diffusion times; shows the spreading of off-diagonal coupling with t.
fig06_precision_eigenbasis.pdf	$\lvert U^\top J_t U\rvert$ heatmaps, with U the eigenbasis of Σ_0, for $t \in \{0.01, 0.30, 1.00, 2.50\}$. Diagonal-dominant only at early t.
fig07_eigenvectors.pdf	Leading 6 eigenvectors of Σ_0; these are essentially the discrete cosine modes of the chain.
fig08_precision_symlog_heatmaps.pdf	J_t on a symmetric-log scale at $t \in \{0.01, 0.15, 0.40\}$, to expose the long-range tail of the off-band entries.
fig09_deterministic_vs_ stochastic_precision.pdf	Stochastic AR(1) precision vs the deterministic-control precision (rank-1 propagation, $a_u = \alpha^u a_0$) at three t. The deterministic case is *not* sparse—a separate fill-in mechanism.
fig10_locality_observables.pdf	Four-panel summary of $\rho_3(t)$, $b_{0.95}(t)$, $\bar{r}(t)$, and neighbour score correlation. The maximum mean range is at $t \approx 1.21$; the minimum tridiagonal mass at $t \approx 0.76$.
fig11_posterior_mean_profile.pdf	Row of the posterior-mean sensitivity $e^{-t}\Sigma_0 J_t$ for five values of t; decays smoothly with frame distance.

Table 2: Gaussian benchmark figure pack (`research_continuation_v2/figures/`). All eleven generated by a single run of `generate_all_figures.py`; reproducible from the recorded summary in `figure_summary.txt`.

Validation snapshot. The closed-form Laplace K=1 score and Hessian used to draw Table 3 are validated to $\sim 10^{-9}$ and $\sim 10^{-8}$ (FD-limited) in Table 1; the centre-curvature Mills-ratio identity is verified to $< 10^{-6}$ by the unit test `test_laplace_K1_centre_curvature` in Section 7.

5 Laplace AR(1) at K=2

Goal. Three-frame chain $a_0 \to a_1 \to a_2$ with two Laplace innovations. The exact joint density of $\boldsymbol{x} = (x_0, x_1, x_2)$ admits a Fourier representation $\hat{p}_t(k)$ with a closed form, but no elementary 1D reduction; $p_t(\boldsymbol{x})$ is recovered by a 3D Fourier inversion (class `LaplaceAR1_K2` in `Experiments/scores_exact.py`).

Source.

- Module: `Experiments/scores_exact.py`, class `LaplaceAR1_K2`
- Driver (main figure pack): `Research/research_continuation_v2/code/generate_all_figures.py` (functions for fig15–fig17)
- Figures: `Research/research_continuation_v2/figures/fig15_*…fig17_*.pdf`

Caveat. **Figures `fig15`, `fig16`, `fig17` use a K=1 proxy for the K=2 score** (see source comments at `generate_all_figures.py:608--622`). The proxy uses `LaplaceAR1_K1.score` on the marginal slice with the "other" frame fixed to zero, instead of the exact `LaplaceAR1_K2.score`

File	What it shows
`fig12_laplace_vs_gaussian_innovations.pdf`	$p_\eta(\eta)$ for Laplace and Gaussian at the same variance, on $\eta \in [-3, 3]$. Heavier tails of Laplace, sharper peak, same standard deviation.
`fig13_k1_score_laplace_vs_gaussian.pdf`	1D slices of $\mathcal{S}_0(x_0, 0, t)$ and $\mathcal{S}_1(0, x_1, t)$ at $t \in \{0.1, 0.5, 1.5\}$. The Gaussian score is linear; the Laplace score saturates as $\lvert x\rvert \to \infty$ (residual-coordinate Mills-ratio behaviour).
`fig14_k1_score_hessian.pdf`	Diagonal entry $-\partial^2_{x_0} \log p_t$ and cross entry $-\partial_{x_0}\partial_{x_1} \log p_t$ along the diagonal $x_0 = x_1$, three values of t; central FD with $h = 2\times10^{-4}$. The Gaussian curves are flat (constant Hessian); the Laplace curves are bumped near zero.
`fig01_innovations_and_kernel.pdf` (audit)	Laplace innovation density and the screened-Poisson Green's kernel that appears in the Fourier representation of p_t.
`fig02_score_slices.pdf` (audit)	Residual score $\psi_t(r)$ along the line $r = x_1 - \alpha x_0$; explicit saturation $\psi_t(r) \to \pm 1/b_t$ as $r \to \pm\infty$.
`fig03_hessian_diagonal.pdf` (audit)	Curvature $\kappa_t(r) = -\psi_t'(r)$ profile, peak at $r = 0$ given by Mills-ratio closed form.
`fig04_hessian_heatmaps.pdf` (audit)	Heatmaps of $(\mathcal{H}_t)_{ij}(\boldsymbol{x})$ on the (x_0, x_1)-plane; the residual line $r = 0$ is visible as a ridge.

Table 3: Laplace K=1 figure pack. All quantities below are matched to the closed-form audit table 1.

File	What it shows
`fig15_k2_score_laplace_vs_gaussian.pdf`	Score components $\mathcal{S}_0$ (vs x_0, x_1=0) and $\mathcal{S}_1$ (vs x_1, x_0=0) at $t \in \{0.3, 0.8\}$, Laplace vs Gaussian.
`fig16_monte_carlo_score_validation.pdf`	Monte-Carlo score estimate (N=5×10^4 samples, Gaussian-kernel density with bandwidth 0.15, central FD on the kernel estimate) vs the closed-form score along $\mathcal{S}_0(x_0, 0, t{=}0.5)$.
`fig17_k2_hessian_band_mass.pdf`	Tridiagonal mass fraction of the empirical Hessian along the diagonal slice $x_0 = x_1$, Laplace vs Gaussian, two values of t.

Table 4: Laplace K=2 figures. See the proxy caveat below.

Fourier inversion. The proxy is qualitatively faithful for the leading and middle components, but a high-fidelity K=2 figure based directly on `scores_exact.py` is on the deliverables list and is *not* yet produced; it is open direction (O1)/(O2) of the synthesis manuscript.

Parameters. Same as Section 4: $\alpha = 0.9$, $b = 0.3082$, $\mu_0 = 0$, $\sigma_0^2 = 1$. Score grids: $x \in [-1.5, 1.5]$, 12 nodes per axis. Monte-Carlo seed: 42.

6 Conjecture 15 grid sweep (Hessian-rank for Laplace K=2)

Statement under test. For the Laplace AR(1) K=2 chain the Hessian $\mathcal{H}_t(\boldsymbol{x}) \in \mathbb{R}^{3\times3}$ of the noised log-density is *approximately rank 2* on the bulk of the support, with the top-2 subspace aligned with the predicted innovation-coupling basis.

Source.

- Notebook (quick run): `Experiments/conjecture15_k2_rank_notebook.ipynb` (executed copy: `conjecture15_k2_rank_notebook.executed.ipynb`)

- Batch driver: `Experiments/conjecture15_overnight_batch.py`
- Medium-resolution suite: `Experiments/conjecture15_grid_suite.py`
- Figures: `figures/conjecture15/` (54 PNG files at quick resolution)
- Note: `Research/conjecture15_test_note.md`

Method (notebook).

1. Build a $5{\times}5{\times}5$ grid on $[-2,2]^3$.
2. At each point, compute $\mathcal{H}_t(\boldsymbol{x})$ by central finite differences of `LaplaceAR1_K2.score` (Fourier inversion with $N = 21$, $K_{\max} = 8$).
3. SVD $\mathcal{H} = U\Sigma V^\top$; record $\sigma_1 \geq \sigma_2 \geq \sigma_3$.
4. Plot σ_3 slice heatmaps, σ_3/σ_1 maps in residual coordinates, and the principal angle between the numerical top-2 subspace $\mathrm{span}(u_1, u_2)$ and a heuristic predicted reference subspace $\mathrm{span}(e_1, e_2 + \alpha e_3)$ (orthonormalised).
5. Repeat with two FD steps ($h = 10^{-2}$ and $h = 5{\times}10^{-3}$) for stability.

Profiles.

- *Quick* (notebook, default): $\alpha \in \{0.3, 0.8, 0.95\}$, $t \in \{0.1, 0.5, 1.5\}$, grid 5^3, Fourier $N = 21, K_{\max} = 8$, two FD steps.
- *Grid suite* (`conjecture15_grid_suite.py`): same α, t lists, grid 9^3 on $[-3.5, 3.5]^3$, Fourier $N = 31, K_{\max} = 10$. Output as JSON+PNG into `figures/conjecture15/grid_suite/`.
- *Paper* (HPC, `HPC_AGENT_HANDOFF_CONJECTURE15.md`): Stage A/B/C with denser grids and the exact predicted basis; *pending*.

Quick-sweep results, $h = 10^{-2}$. The three primary indicators per configuration are the median and 95th percentile of σ_3/σ_1 over the grid, and the mean principal angle in degrees to the heuristic predicted basis.

Table 5: Conjecture 15 quick-sweep results (`ratio_median` = $\mathrm{med}(\sigma_3/\sigma_1)$, `ratio_q95` = $Q_{0.95}(\sigma_3/\sigma_1)$, `angle_mean_deg` = mean principal angle in degrees).

α	t	ratio_median	ratio_q95	angle_mean_deg
0.30	0.1	0.0385	0.4120	49.83
0.30	0.5	0.3621	0.6900	56.40
0.30	1.5	0.8987	0.9214	59.64
0.80	0.1	0.0585	0.2272	69.92
0.80	0.5	0.2954	0.3853	72.59
0.80	1.5	0.7958	0.8134	77.16
0.95	0.1	0.0747	0.1798	74.22
0.95	0.5	0.2492	0.3152	77.78
0.95	1.5	0.7535	0.7712	80.21

Stability ($h = 10^{-2}$ vs. $h = 5{\times}10^{-3}$). Relative differences in σ_3/σ_1 are between $\sim 10^{-7}$ and $\sim 10^{-4}$ on the median, and between $\sim 10^{-6}$ and $\sim 4{\times}10^{-3}$ on the 95th percentile. The numerics at the quick resolution are internally stable.

Caveat. **The quick run is regime-dependent and does *not* support a machine-precision rank-2 claim** (verbatim verdict in `Research/conjecture15_test_note.md`). At small t and large α the third singular value is small in median; at $t = 1.5$ it is comparable to σ_1 across the entire sweep, so the conjecture is at most an *approximate, regime-dependent low-rank* statement. The grid 5^3 and Fourier $N = 21$ are intentionally coarse; a paper-grade run with denser grid, larger Fourier resolution, and the exact predicted basis is pending HPC (`HPC_AGENT_HANDOFF_CONJECTURE15.md`).

Figure inventory. At quick resolution: 9 configurations × 3 slice files each × 2 FD steps = 54 PNG files in `figures/conjecture15/`, named

- `sigma3_slices_alpha*_t*_h*.png`
- `ratio_map_alpha*_t*_h*.png`
- `angle_map_alpha*_t*_h*.png`

The grid-suite (9^3) regenerates the same figure types into `figures/conjecture15/grid_suite/`.

7 Self-tests in `scores_exact.py`

The module `Experiments/scores_exact.py` packages the closed-form score code used by Sections 5 and 6 and includes six unit tests, run by executing the module as `python scores_exact.py`.

Table 6: Self-tests in `scores_exact.py`.

Test	What it asserts	Tolerance
`test_gaussian_ar1_kalman_vs_matrix`	For Gaussian AR(1), Kalman-smoother score and matrix-form score $-J_t(\boldsymbol{x} - e^{-t}\mu)$ coincide; sweep $K \in \{2, 4, 8, 16, 32\}$, $\alpha \in \{0, 0.3, 0.7, 0.95\}$, $t \in \{0.05, 0.3, 1, 3\}$.	$< 10^{-10}$
`test_laplace_K1_mc`	The residual-coordinate score $\psi_t(r)$ saturates to $\pm 1/b_t$ as $\lvert r\rvert \to \infty$ (here at $r = \pm 30$).	$< 10^{-6}$
`test_laplace_K1_centre_curvature`	Mills-ratio closed form for $\kappa_t(0)$ agrees with direct second derivative of the closed-form score; four t.	rel. $< 10^{-6}$
`test_laplace_K2_fourier_density_positive`	Fourier-inverted $p_t(\boldsymbol{x})$ at the chain mean is positive and finite; two t.	—
`test_laplace_K2_score_finite_diff_is_stable`	The K=2 score at the mean is finite (FD-stable).	—
`test_gaussian_limit_of_laplace_K1`	At t=5 the Laplace K=1 curvature $\kappa_t(0)$ approaches the Gaussian limit $1/(\tau^2 + 2b_t^2)$.	rel. $< 5\%$

Status. All six tests pass on the current code (logged in `CLAUDE_CODE_TASKS.md`).

8 Cross-reference table

9 Reproducibility

From the project root `Score_Diffusion/`:

```
# 1. Audit suite (must pass first):
```

Table 7: Where each experiment is used in the synthesis manuscript `merged_core_notes.tex`.

This document	Synthesis section	Synthesis label
Section 2	§4 (Gaussian benchmark, §5 Laplace K=1)	—
Section 3	§4 numerical figure suite	"fig01–11" Remark
Section 4	§5.5 (Result 5.2 illustration), §5.6 (Result 5.3)	—
Section 5	§6 (Result 6.4 / Result 6.7), Open dir. (O1)+(O2)	—
Section 6	§9 (Conjecture 9.1)	"Current numerical status"
Section 7	§8 implementation note	—

```
cd Research/laplace_ar1_audit/code
python3 run_checks.py           # writes check_report.txt

# 2. Gaussian + Laplace K=1, K=2 figures (fig01--fig17):
cd ../../research_continuation_v2/code
python3 generate_all_figures.py  # writes ../figures/fig*.pdf
                                 # and code/figure_summary.txt

# 3. Audit illustration figures (Laplace K=1 only):
cd ../../laplace_ar1_audit/code
python3 make_figures.py          # writes ../figures/fig01--04.pdf

# 4. scores_exact self-tests:
cd ../../../Experiments
python3 scores_exact.py          # six unit tests

# 5. Conjecture 15 quick sweep (notebook):
jupyter nbconvert --to notebook --execute \
    conjecture15_k2_rank_notebook.ipynb \
    --output conjecture15_k2_rank_notebook.executed.ipynb

# 6. Conjecture 15 medium sweep (script):
python3 conjecture15_grid_suite.py
```

All scripts are deterministic except for the Monte-Carlo step in `fig16`, which uses `numpy.random.seed(42)`.

10 Open numerical work

- *High-fidelity Laplace K=2 figures.* Replace the K=1 proxy in `fig15--17` by direct calls to `LaplaceAR1_K2.score`; plot off-diagonal Hessian bump location vs. α, t (Open dir. O2 of the synthesis).
- *Conjecture 15 paper-grade run.* Stages A/B/C of `HPC_AGENT_HANDOFF_CONJECTURE15.md`: denser grid, exact innovation-coupling basis, extended $\alpha \times t$ sweep. Verdict slot reserved in `CLAUDE_CODE_TASKS.md`.
- *Architectural experiment* (Track B of `CLAUDE_CODE_TASKS.md`). Empirical test of the posterior-factor-graph parameterisation against a black-box neural baseline; benchmark ladder defined in `Experiments/posterior_chapter.md` §8.
- *Stationarity-aware estimator* (Open dir. O6 of the synthesis): fold the AR(1) stationarity prior into the score-matching loss and measure the variance reduction.